

MA3831 Assignment 3 - Capstone Project

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Introduction

Predicting the stock market is a topic of great interest to many people. News and consumer sentiment play a crucial role in influencing stock prices, as traders continuously monitor financial news and social media to assess supply, demand, and market trends. This process requires constant tracking, interpretation, and decision-making to respond effectively to market fluctuations. It is therefore evident that an NLP system could be of great use, being able to (1) summarise articles and identify key information and (2) provide sentiment variables as input features for a stock price prediction model.

This report presents a pipeline that collects news articles using web scraping techniques and processes them through NLP systems to extract meaningful insights. All code and reporting files are available on GitHub.

Web Crawler

Data Type Selection

It is clear that a textual data corpus is required for the exploration of NLP techniques in this setting. There are numerous sources on the world wide web that were considered.

The first source that was considered was the ASX market announcements [1]. The announcements are straight from the source, come at a high frequency, but come with the drawback of sometimes being mundane and not helpful, the format being in a pdf document which is not conducive to text mining with tool such as `BeautifulSoup`, and it being against the terms of use of the ASX [2].

The second source of data that was considered was social media platforms, in particular Twitter and Reddit. Using social media platforms would afford a large quantity of data, but with each document itself being small. Using social media also has the drawback of being unstructured, and also the text data using colloquial language and having spelling mistakes, with extensive preprocessing required as found in many papers [put the papers here, say something about how social media users aren't held to a standard to be relied upon for factual or accurate information](#)

In contrast, financial news articles are more factual than social media, with the added benefit of being longer in length, having correct spelling and grammar. Furthermore, websites can offer more structure in the way they present their articles, with easy ways to scrape key metadata, [perhaps reference the next section where this is elaborated on](#).

Website Selection

Table 1 shows the comparison of various websites that contain financial and market-related articles, evaluating their suitability for scraping.

The smallcaps website was selected for the following reasons: - no dynamic content, so no elaborate web scraping web driver is required, can just use `BeautifulSoup` - the website is all about stocks, so no need to filter out non-stock related articles - well labelled articles with stock codes

Website Name (URL hyperlinked)	Scraping Permitted in Terms of Use?	Allows Article Scraping in Robots.txt?	Uses Cloudflare?	Notes of quality
Market Index	No	Yes	No	Company tags on articles
Yahoo Finance	No	No	No	no mentioned stock labelling. General politics also included which is difficult to relate to individual stocks
Trading Economics	Yes	Yes	No	General politics and economics also included — would require filtering (unstructured)
Financial Review	No	No	No	
The Motley Fool	No	Yes	No	
Sydney Morning Herald	No	No	No	
The Market Online	No	Yes	No	
ABC News	Yes	Yes	No	Regular news mixed with financial news, difficult to use, untagged stock market
Small Caps	Yes	Yes	No	Sitemap highly conducive to scraping. Stock codes provided in articles. All about stocks — no general politics. Static web content (use BeautifulSoup)
Morning Star	No	Yes	No	No sitemap provided
Australian Stock Report	Yes	Yes	No	Unstructured in comparison to other article sites
Stock Head	Yes	Yes	No	Great sitemap but no list of stock codes in article
The Australian	No	No	No	
Hot Copper	No	Yes	No	

Table 1: Comparison of financial and market-related websites for article scraping suitability. Green rows indicate favourable conditions for scraping.

there is however drawbacks to this particular website and especially only choosing one website from which to scrape from.

asdf

Web Scraping

To begin this analysis, we scrape the sitemap of `smallcaps.com.au`. This offers the benefit of having a consolidated list of all article URLs, eliminating the need to emulate human navigation through pages of website search results using `Selenium`. The `lxml` parser is used for this task due to its fast performance and capability to handle both XML and HTML scraping. table 2 shows the distribution of the last modified date of the urls in the XML sitemap.

Article Year	Number of Observations
2023	10,468
2024	2,374
2025	323
Total	13,165

Table 2: Number of Observations per Year

Now, individual article content needs to be scraped. The following listing uses `BeautifulSoup` and the `requests` library to extract the contents of a given URL, which was sourced from the sitemap. The `lxml` parser was chosen again for its speed, and since the necessary external C dependency had

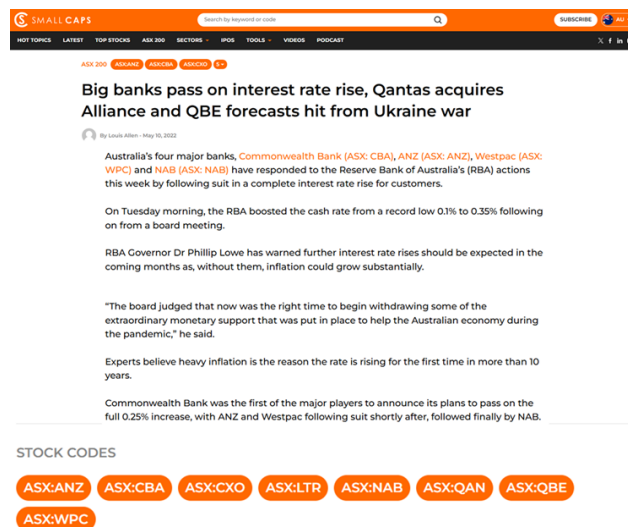


Fig. 1. Example Article [3], containing fields such as title, author, date, and stock codes

already been installed with the `lxml-xml` parser during the XML sitemap scraping, no additional setup was required [4]

The contents of the HTML were analysed and parsed into a dictionary capturing the title, author, date, sector, stock codes, and the article body, as demonstrated in the listing below. `BeautifulSoup` was used to extract data from the relevant tags. Below is an example output from the parsing process:

Listing 1: Example HTML extraction code use and dictionary response

```
1 soup_test = extract_html_from_url('https://smallcaps.com.au/galan-lithium-directors-
  show-confidence-share-purchase/')
2 parsed_dict = parse_html_to_dict(soup_test)
3 parsed_dict
4
5 {'title': 'Galan Lithium directors show confidence with $1.4 million share purchase'
  ,
6  'author': 'Colin Hay',
7  'date_string': 'May 23, 2024',
8  'sector': 'Mining',
9  'stock_codes': ['ASX:GLN'],
10 'document': 'Galan Lithium (ASX: GLN) managing director Juan Pablo....}'
```

Machine Learning

After having scraped the articles for the documents and sentiments? a machine learning model was developed to predict stock price. figs. 2 and 3 show the structure of a typical feed forward neural network and a recurrent neural network. Recurrent neural networks (RNNs), and others like RNNs, are particularly well suited to sie seues data

[5] show that an LSTM model, when trained on timeseries price data and sentiment data from news articles, can outperform a model trained on price data alone. Analysis of the literature suggests that SVM models, whilst the most popular [6] often underperformed in a financial setting, due to parameter optimisation being required [6, 7]

Talk about the adam optimiser According to Kingma et al., 2014, the method is "computationally

efficient, has little memory requirement, invariant to diagonal rescaling of gradients, and is well suited for problems that are large in terms of data/parameters". [8,9]

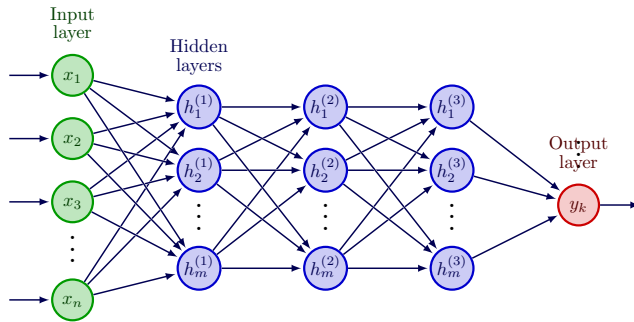


Fig. 2. Typical "FeedForward" Neural Network Topology [10]

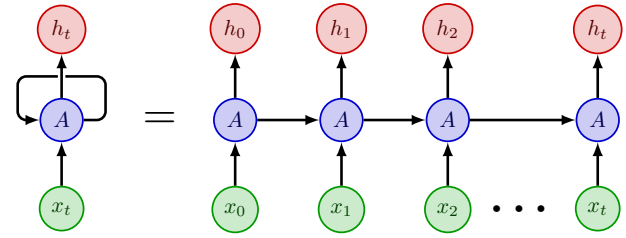


Fig. 3. Recurrent Neural Network Topology [11]

the events extracted from the news are usually quite sparse, resulting in unreliable and unsustainable predictive power [7] (bar for bar)

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